

Potential Talents - Technical Report

Candidate ranking with NLPL Model 40 Word2Vec, transparent relevance scoring, and recruiter feedback

0.948 NDCG@10 - Aspiring HR	0.935 NDCG@10 - Seeking HR	100% Token occurrence coverage	52 Unique titles evaluated
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Executive Summary

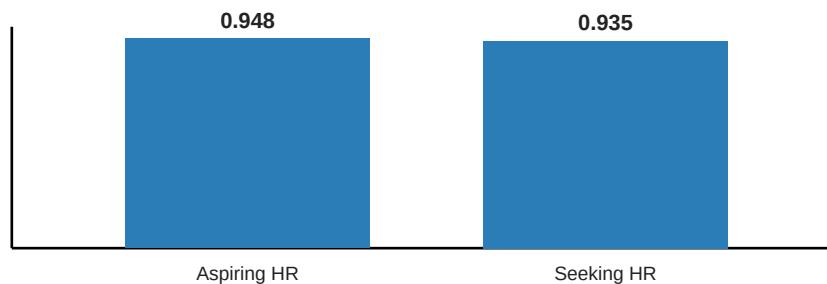
The final workflow uses **NLPL Model 40 only**. Duplicate titles are collapsed before evaluation, a transparent rule score supplies graded relevance, Model 40 ranks titles by cosine similarity, and recruiter feedback shifts the query toward a starred candidate. The verified run achieved NDCG@10 of **0.948** and **0.935**, with complete vocabulary coverage.

Project Objectives

Overall objective: Develop and evaluate a transparent, reproducible candidate-ranking workflow that prioritizes Human Resources candidates for the two recruiter searches using NLPL Model 40 Word2Vec, rule-based relevance, and recruiter feedback.

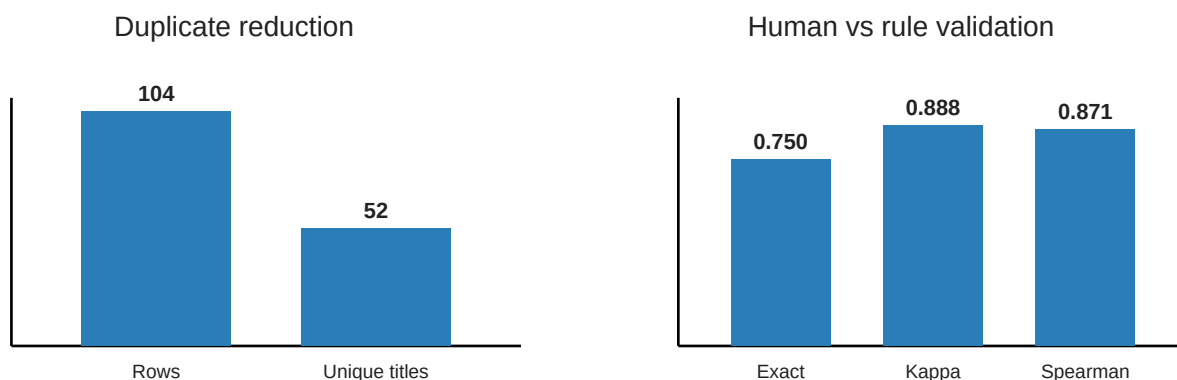
- 1. Data audit and preparation.** Deduplicate 104 records to 52 unique titles and characterize the title vocabulary.
- 2. Auditable relevance benchmark.** Define occupational relevance and intent alignment using $R = H(0.70 + 0.30I)$.
- 3. Independent validation.** Compare rule levels with manually assigned 0-3 human grades.
- 4. Semantic Stage-1 ranking.** Rank Model-40 mean embeddings by cosine similarity.
- 5. Ranking evaluation.** Measure NDCG@10 and vocabulary/OOV coverage for both queries.
- 6. Recruiter feedback evaluation.** Quantify the personalization-stability trade-off across 34 strong scenarios and translate results into guidance.

Model 40 Stage-1 ranking quality



1. Data Audit and Relevance Reference

The source contains 104 candidate rows but only 52 unique job titles. Ranking each unique title once prevents repeated text from receiving disproportionate weight. Fourteen titles repeat and the largest exact-title frequency is seven.



Rule-based relevance

The query-specific reference is $R = H(0.70 + 0.30I)$. H is occupational relevance and I is intent alignment. Because intent is multiplied by occupational relevance, generic seeking-employment wording cannot become HR-relevant on intent alone. Clear employer solicitations receive zero candidate relevance.

Validation measure	Result
Exact ordinal agreement	39/52 (75.0%)
Quadratic weighted Cohen's kappa	0.888
Spearman correlation	0.871

After deduplication, 31/52 titles explicitly contain Human Resources, 34/52 contain direct HR evidence, 12 contain aspiring, and 10 contain seeking. This supports an auditable rule layer plus semantic ranking.

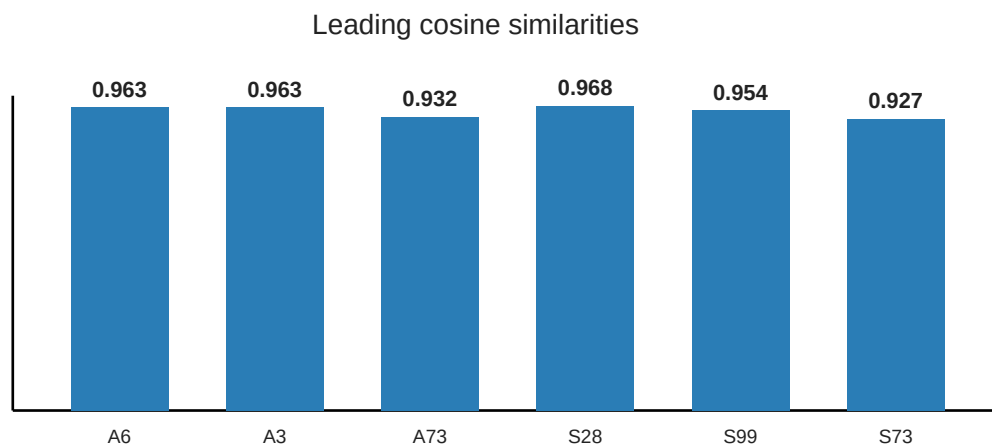
2. Model 40 Semantic Ranking

Model: **NLPL model 40 - English CoNLL17 Word2Vec 100D, continuous skip-gram**. Vocabulary header: 4,027,169; official archive approximately 3.03 GB.

Coverage quantity	Result
Meaningful title-token occurrences	368
Title + query occurrences	374
Resolved token occurrences	374 (100%)
Unique meaningful tokens	208
Unique OOV tokens	0

Stage-1 results

NDCG@10 is **0.948** for aspiring HR and **0.935** for seeking HR, mean **0.942**. The leading aspiring IDs are 6, 3, 73; the leading seeking IDs are 28, 99, 73.

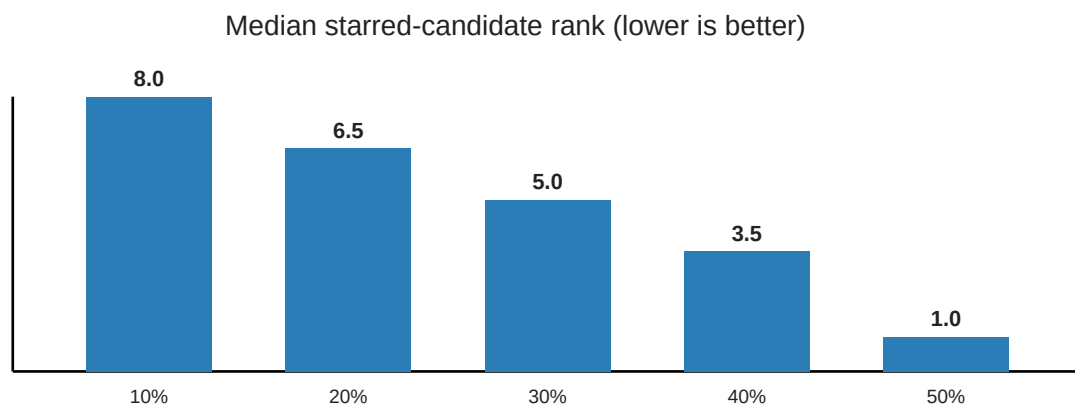


NDCG@10 values close to 1.0 indicate that the most relevant titles appear near the top of the semantic ranking. Model 40 achieved complete token coverage for the analysis corpus.

3. Recruiter Feedback and Reranking

A starred candidate shifts the query as $q_2 = \text{normalize}((1-w)q + w d^*)$. The experiment evaluates 34 strong title/query scenarios at five feedback weights.

Feedback	Median starred rank	Median top-10 retained
10%	8.0	9/10
20%	6.5	9/10
30%	5.0	9/10
40%	3.5	8/10
50%	1.0	8/10



Interpretation

No tested weight satisfies the former joint target of median starred rank ≤ 2 while retaining at least 9/10 of the original top ten. Model 40 therefore does **not** claim a universal optimum. At 30%, both representative starred candidates move 2 -> 1 while retaining 9/10 and 10/10 of the original top ten.

Query / candidate	Rank	Top-10 retained	NDCG@10
Aspiring HR - ID 3	2 -> 1	9/10	0.948 -> 0.955
Seeking HR - ID 99	2 -> 1	10/10	0.935 -> 0.940

4. Practical Recommendation and Limitations

Use the rule framework as an auditable relevance screen, Model 40 as the semantic ordering mechanism, and recruiter feedback as a controlled personalization layer. Avoid a universal cosine-similarity cutoff because similarity is query- and pool-dependent.

Limitation	Implication
Mean pooling	Long titles can dilute important HR terms.
Limited context	Word2Vec cannot fully resolve directionality or sentence meaning.
Title-only feature	Skills and experience absent from the title cannot affect ranking.
Rule portability	New occupations require new role/intent definitions and validation.
Feedback strength	Higher weight moves starred candidates up faster but disrupts more of the shortlist.

Reproducibility

Notebook 01 runs from the included CSV files. Notebooks 02 and 03 require the official NLPL **40.zip** archive and stream only the vectors needed for the 52 titles and two queries. Official archive: <https://vectors.nlpl.eu/repository/20/40.zip>